

SOCIAL DESIRABILITY BIAS IN CATI, IVR, AND WEB SURVEYS

THE EFFECTS OF MODE AND QUESTION SENSITIVITY

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Abstract Although it is well established that self-administered questionnaires tend to yield fewer reports in the socially desirable direction than do interviewer-administered questionnaires, less is known about whether different modes of self-administration vary in their effects on socially desirable responding. In addition, most mode comparison studies lack validation data and thus cannot separate the effects of differential nonresponse bias from the effects of differences in measurement error. This paper uses survey and record data to examine mode effects on the reporting of potentially sensitive information by a sample of recent university graduates. Respondents were randomly assigned to one of three modes of data collection—conventional computer-assisted telephone interviewing (CATI), interactive voice recognition (IVR), and the Web—and were asked about both desirable and undesirable attributes of their academic experiences. University records were used to evaluate the accuracy of the answers and to examine differences in nonresponse bias by mode. Web administration increased the level of reporting of sensitive information and reporting accuracy relative to conventional CATI, with IVR intermediate between the other two modes. Both mode of data collection and the actual status of the respondent influenced whether respondents found an item sensitive.

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Introduction

Questions about sensitive topics are common in household surveys. Questions can be considered sensitive if respondents perceive them as intrusive, if the questions raise fears about the potential repercussions of disclosing the information, or if they trigger social desirability concerns (Tourangeau and Yan 2007). In this paper we focus on the third type of sensitive questions, those that trigger social desirability concerns. The concept of social desirability rests on the notions that there are social norms governing some behaviors and attitudes and that people may misrepresent themselves to appear to comply with these norms. In the general population, for example, voting is often seen as a civic duty and not voting as constituting a violation of the norm. Thus some survey respondents overreport voting (Belli, Traugott, and Beckmann 2001). Similarly, some respondents underreport undesirable behaviors, such as illicit drug use or heavy drinking (see Tourangeau and Yan 2007). A wide variety of topics may be prone to social desirability effects, ranging from abortion (Jones and Forrest 1992) to religious service attendance (Presser and Stinson 1998) to having a library card (Parry and Crossley 1950), though some “sensitive” topics are doubtless more susceptible to social desirability biases than others.

Numerous methodological studies have established that self-administration lessens social desirability effects. For example, Tourangeau, Rips, and Rasinski (2000) found that every one of nine mode experiments measuring self-reported illicit drug use showed higher rates of reporting with self-administration than with interviewer administration.

Although this finding on the impact of self-administration has been replicated many times, it is not clear whether computerized self-administration is more effective at reducing social desirability bias than traditional paper methods (see Tourangeau and Yan 2007, for a recent, somewhat inconclusive, meta-analysis). Indeed, there are few studies that compare different methods of self-administration, e.g., paper self-administered questionnaires versus audio-computer-assisted survey instruments versus interactive voice recognition versus Web administration (see Tourangeau and Smith 1996 for an exception).

Another limitation of the prior literature on mode differences in socially desirable responding is that most of the studies lack validation data. Such data are often difficult (or impossible) to obtain; when they are available, they may involve very specialized populations. The lack of validation data forces investigators to make two assumptions in determining which mode leads to “better” results. The first assumption is that social desirability concerns lead respondents to underreport socially undesirable behaviors so that the data collection mode that yields higher levels of reporting is the more accurate one. The second assumption is that lower reports of socially desirable behaviors reflect more accurate answers. The extent to which these assumptions are correct cannot be determined without validation data.

Validation data are also useful in two other ways. First, underreporting of socially undesirable behavior is partly a function of perceived question sensitivity but question sensitivity depends on the respondent's actual status on the variable in question. For example, an inquiry about voting is subject to social desirability effects only among those respondents who did not vote. Thus, external validation data can permit more sensitive tests of the effects of data collection mode (and of other variables that might affect reporting accuracy) by focusing on those who are at risk of misreporting. Second, mode comparisons, like other comparisons, are subject to nonresponse. If respondents drop out after the assignment to mode and the dropout rates differ by mode, nonresponse bias can affect the results; mode differences then represent the combined effect of differences in nonresponse bias and differences in reporting error. External data for both respondents and nonrespondents allow a direct assessment of the nonresponse bias. Finally, most studies of social desirability have focused either solely on socially undesirable behavior or socially desirable behavior, with the great majority examining undesirable behavior. As a result, we know little about the relative effect of different modes of data collection on the two forms of misreporting error produced by social desirability.

We address all these issues in the study reported here. First, the study includes questions prone to both over- and underreporting. Second, it compares interviewer administration via computer-assisted telephone interviewing (CATI) with two methods of self-administration, interactive voice response (IVR) and the Web, both of which have become increasingly widespread. Third, it uses external records to determine respondents' true values. In addition, the study collected ratings of item sensitivity at the end of the survey. Most prior studies of social desirability biases in survey reporting rely on the researchers' judgments about which items are sensitive (although see Bradburn, Sudman, and Associates 1979, and Holbrook, Green, and Krosnick 2003, for two exceptions). It is useful to have respondent data to confirm or disconfirm those judgments.

Our data come from the Joint Program in Survey Methodology 2005 Practicum survey. We use those data to explore the following questions:

1. What are the relative effects of the two self-administered modes and how do they compare to interviewer administration of the questionnaire? Are there differences in the mode effects for socially desirable and undesirable behaviors?
2. Do the record check data support the usual assumptions regarding the direction of errors (that is, most errors are in the socially desirable direction)? Are any other sources of systematic error apparent in the reports?
3. Does the perceived level of question sensitivity vary with mode of data collection? If so, are mode differences mediated by perceptions of an item's sensitivity?

The Study

Each year, the Joint Program in Survey Methodology (JPSM) at the University of Maryland carries out a survey in conjunction with its Practicum class. The 2005 JPSM Practicum involved a survey of University of Maryland alumni. Interviewing was conducted by Schulman, Ronca, and Bucuvalas, Inc., in August and early September, 2005. The alumni were initially contacted by telephone and were asked a brief set of screening questions about their personal and household characteristics (including access to the Internet). The survey introduction did not refer to the potentially sensitive items in the questionnaire:

Hello, my name is [INTERVIEWER'S NAME] and I'm calling on behalf of the University of Maryland. You have been randomly selected to participate in a survey of Maryland alumni. I'm not calling to ask you for a donation. I will be asking you about your college experience, interest in alumni activities, and community involvement. Your participation is strictly voluntary and you may skip any questions you don't want to answer. All of your responses will be kept confidential. The survey will take about 10 minutes.

ITEMS USED

The main questionnaire included 37 questions about educational experiences, current relationships with the University, community involvement, and a final set of questions to measure the sensitivity of some of the earlier items. Most of the questions were included to further the research aims of the Alumni Association and are not relevant here. The items used for this study are listed in table 1 and in the remaining part of this section.

Table 1 lists the items (and the question wording of the items) for which records were available from the Registrar's office or the Alumni Association. Three of them involve socially undesirable behaviors—dropping a class, getting an unsatisfactory grade, and receiving an academic warning or being placed on probation. Four others asked about socially desirable behaviors—receiving academic honors, being a member of the Alumni Association, and donating money to the University of Maryland (two items asked about this last topic). The eighth item asked about the respondent's grade point average (GPA). We treat GPAs lower than 2.5 as undesirable and those higher than 3.5 as desirable. A GPA of 3.5 or more qualifies for the Dean's list at the University of Maryland, whereas GPAs below 2.5 are poor—less than 2.0 triggers academic warning.¹

It is useful to note that the behaviors tapped by these items vary in their actual prevalence. According to the records, fewer than 3 percent of all alumni had been placed on academic warning or probation. This stands in contrast to the

1. The perception of what constitutes a good and bad GPA will vary across individuals. However, this is true for the classification of any item. Using cut-points that reflect formal aspects of the academic system seemed a reasonable approach to capturing typical views.

Table 1. Sensitive Items from the 2005 JPSM Practicum and Prevalence in Sampling Frame

Item	Question	Frame
Dropping a class	During the time you were an undergraduate at the University of Maryland, did you ever drop a class and receive a grade of “W”?	70.9%
Receiving an unsatisfactory grade (D or F)	Did you ever receive a grade of “D” or “F” for a class?	62.7%
Receiving an academic warning/probation	Were you ever placed on academic warning or academic probation?	2.6%
GPA	What was your cumulative overall undergraduate grade point average or GPA at the time you received your undergraduate degree?	<2.5 = 15.2%
		>3.5 = 18.6%
Receiving academic honors	...did you graduate with <i>cum laude</i> , <i>magna cum laude</i> , or <i>summa cum laude</i> ?	9.5%
Being in the Alumni Association	Are you a dues-paying member of the University of Maryland Alumni Association?	7.3%
Donating money to UMD	Since you graduated, have you ever donated financially to the University of Maryland?	25.3%
	Did you make a donation to the University of Maryland in calendar year 2004?	8.4%

almost two-thirds of the alumni who had received an unsatisfactory or failing grade and the roughly 70 percent who had dropped a class at least once during their time as an undergraduate at Maryland.

These items came relatively early in the questionnaire and were followed by questions on topics such as the respondent’s current relationship with the University, views about alumni activities, attendance at reunions, and donations to the Alumni Association. The final item in the questionnaire asked respondents to rate the sensitivity of some of the earlier items. It began

Questions sometimes have different effects on people. We’d like your opinions about some of the questions in this interview. [CATI/IVR:] As I mention a question, please indicate whether you think it might make people you know falsely report or exaggerate their answers. [Web:] Do you think that the following questions might make people you know falsely report or exaggerate their answers? (Please answer “yes” if you think a question might make people falsely report or exaggerate their answer. Otherwise please answer “no”).

We used this item to assess the sensitivity of the first four questions in table 1 (e.g., What about the question on dropping a class and a receiving grade of W? Would you say that this question might make people you know falsely report or exaggerate their answer?).²

SAMPLING AND RESPONSE RATES

A random sample (stratified by graduation year) was drawn from the 55,320 individuals listed in the Registrar's records as receiving undergraduate degrees from the University of Maryland from 1989 to 2002. Of the 20,000 graduates sampled, 10,325 could be matched to Alumni Association records containing telephone contact information. After various ineligible numbers were dropped, including those used in pretesting, 7,591 phone numbers were fielded for the survey.³ More than a third of the telephone numbers on the Alumni Association records were invalid (e.g., the number was disconnected or the sample member did not live at the residence linked to the number). In addition, the status of 25 percent of the telephone numbers could not be determined (e.g., an answering machine picked up during every one of the call attempts). A total of 1,501 alumni completed the screener and were randomly assigned to a mode of data collection.⁴ The response rate (AAPOR Response Rate 1; AAPOR 2008) for the screener was 31.9 percent.⁵ Most of the nonresponse appeared to reflect our inability to contact the alumni rather than their unwillingness to complete the interview. The refusal rate is only about 9 percent of the fielded phone numbers excluding the noneligible cases (see table 2).

It is possible that the relatively low response rate caused nonresponse bias. If the participating alumni tended not to exhibit any undesirable behaviors, our mode comparison would be unlikely to show large differences in reporting error for those variables. The random assignment to a mode of data collection occurred after the bulk of the nonresponse, however, and thus nonresponse bias (at least through the screening interviews) would be expected to affect all three mode groups in the same way.

2. One should note that the IVR respondents could not return to previous items and change their answers, though they could skip a question or get it repeated. CATI and Web respondents could conceivably have gone back to the earlier questions, though we are unaware of whether any actually did this.

3. Among the 10,325 graduates for whom phone numbers were available, 14 were listed as residents of Puerto Rico, the Virgin Islands, or on military bases and were excluded from the sample, as were 151 listed with the same phone number as another graduate (in these cases, one graduate was dropped randomly). Also excluded were those who had previously been contacted as part of a small-scale pretest conducted during the Spring of 2005 or a larger-scale, operations test ($n = 1,975$) conducted on July 27–28, 2005.

4. For 37 individuals without access to the Web, the random assignment was restricted to CATI versus IVR.

5. This is a conservative estimate, since it assumes that all cases for which we could not determine eligibility were eligible.

Table 2. Final Disposition Codes

	Total	Percentage	Percentage
Seemingly usable phone numbers fielded	7,591	100.0	—
Not eligible or deceased	2,889	38.1	—
Eligible cases and unknown eligibility	4,702	61.9	100.0
Unknown eligibility	1,914	—	40.7
Eligible, no interview			
Language barrier	33	—	0.7
Physically/mentally unable	7	—	0.1
Noncontact	797	—	17.0
Refusal	441	—	9.4
Partial screener completion	9	—	0.2
Screener completed and assigned to mode	1,501	—	31.9

Table 3. True Status in Percent for Frame Data and Screener Respondents

	Frame data (after successful match to Alumni Association records)	Respondents assigned to mode experiment (95% confidence intervals)
Undesirable characteristic		
GPA < 2.5	15.2	13.7 (12.0–15.5)
At least one D or F	62.7	61.0 (58.5–63.4)
Dropped a class	70.9	69.3 (67.0–71.6)
Warning or probation	2.6	2.7 (1.8–3.5)
Desirable characteristic		
GPA > 3.5	18.6	20.9 (18.9–23.0)
Honors	9.5	11.3 (9.7–12.9)
Donations ever	25.3	38.0 (35.5–40.4)
Donations in 2004	8.4	15.1 (13.2–16.9)
Alumni Association member	7.3	14.5 (12.7–16.3)
Total	10,325	1,501

The availability of record data for both the respondents and nonrespondents allowed us to assess the effects of nonresponse and sampling error on the composition of the sample that was successfully screened. Table 3 shows that there was no relation between screener nonresponse and the true values for any of the socially undesirable behaviors but there was a relation for the socially desirable characteristics—the screener respondents were a little more apt than the frame population as a whole to have performed well as undergraduates and more likely to be members of the Alumni Association and to have donated to the University. We suspect that this reflects the greater accuracy of the contact information for alumni who are donors or are members of the Alumni

Table 4. Dropout Rates by Mode of Data Collection

	Total	CATI	Web	IVR
Initially assigned	1,501	338	639	524
Agreed to participate in Web		—	617	—
Agreed to participate in IVR		—	—	508
Started the main questionnaire	1,107	329	368	410
Number of completes (in percent)	1,003 (66.8)	320 (94.7)	363 (56.8)	320 (61.1)

Association. Even though individuals with these positive characteristics were overrepresented among the screener respondents, they still constituted a minority of respondents, and thus the majority of the respondents were in a position to overreport these behaviors.

Nonresponse occurring after the screening interview could also introduce bias and affect comparisons across the modes of data collection. Table 4 shows the number of cases assigned to each mode of data collection, the number who started the questionnaire, and the number who completed it.⁶ The overall completion rate among the screener respondents was 67 percent.⁷ The rate was highest among those assigned to the interviewer-administered interview (320 completed the main questionnaire out of the 338 assigned to CATI, or 94.7 percent) and lower for the Web group and those switched to IVR, the interactive voice recognition system (56.8 and 61.1 percent, respectively). These differences are highly significant— $\chi^2 = 155.0$ with 2 df ($p < .001$). Most of the nonresponse to the main questionnaire in the Web and IVR conditions occurred before the sample alumni started the questionnaire. Of the 617 respondents who agreed to do the survey over the Web, 40.4 percent (249 of 617) never actually started the main questionnaire; only five Web cases quit partway through. Similarly, most of the IVR cases that didn't complete the main questionnaire (a total of 204 alumni) either refused to be switched to the IVR questionnaire (16 cases) or dropped out during the switch to the automated system (98 more cases). Once respondents began either the IVR or Web survey, IVR cases were more likely to drop out (90 of 410) between the beginning and the end of the interview than the Web cases (5 of 368); the difference between

6. The sample was divided into 40 replicates that were released in sequence to allow changes in the mode allocation throughout the data collection period. The initial goal was to achieve 360 completed cases from each mode. Based on response rate estimates by mode from the operations pretest, the initial allocation of sample cases was set at 19, 56, and 25 percent for CATI, Web, and IVR, respectively. The allocation by mode was adjusted to 24, 36, and 40 percent at the end of August. In the second week of September, all the remaining cases were designated to the IVR mode of collection.

7. This leads to an overall response rate of 21.3 percent (AAPOR Response Rate 1 computed by multiplying the screener completion rate of 31.9 percent times the completion rate after initial assignment of 66.8 percent).

Table 5. True Status (in Percent) for Item Responders, by Item and Mode Group

	CATI	IVR	Web	χ^2 (df; <i>p</i> -value)
Undesirable characteristic				
GPA < 2.5	10.8	9.6	14.6	4.16 (2; .125)
At least one D or F	61.0	58.6	62.3	0.96 (2; .617)
Dropped a class	68.0	64.4	70.7	3.05 (2; .217)
Warning or probation	1.9	2.9	2.2	0.68 (2; .714)
Desirable characteristic				
GPA > 3.5	22.2	23.3	20.8	0.59 (2; .743)
Honors	12.4	15.1	9.9	4.13 (2; .127)
Donations ever	40.1	38.6	42.7	1.19 (2; .552)
Donations in 2004	28.3	30.8	30.4	0.20 (2; .905)
Alumni Association member	16.1	14.4	17.5	1.17 (2; .556)

these dropout rates was significant ($\chi^2 = 65.8$, 1 df, $p < .001$). Overall, there are 1,003 completes that were used in the analyses.

Table 5 shows the distribution by mode of the true statuses according to University records for the responding cases. None of the differences across the three groups is significant, suggesting that the differential nonresponse bias does not jeopardize the mode comparisons.⁸

In addition, there appears to be no systematic relation between the content of the questions and the decision to drop out among those who started the survey. Figure 1 shows where partial respondents dropped out during the main interview by mode. In the Web survey, only three respondents discontinued the survey after the first set of questions (about the year of graduation and their living situation during their college years) and none thereafter. CATI respondents show a similar pattern; 10 respondents dropped out before the first block of sensitive questions and five immediately thereafter. IVR respondents dropped out throughout the questionnaire, a pattern suggesting that fatigue (or annoyance with IVR) was the main cause of breaking off rather than question sensitivity. The rate of breaking off was highest toward the end of the IVR interview, with no jump in breakoffs after the sections with the sensitive items.

To summarize, the response rates to the main questionnaire differed significantly by mode (with CATI having the highest response rate) but the bulk of the nonresponse occurred before respondents started the questionnaire. Across the items for which validation data were available, the respondents in the three mode groups included a similar mix of cases in the desirable and undesirable categories on our key variables. Overall, those with poor academic records or

8. We also tested for differences in age (both raw values and Box-Cox transformed values) as well as year of graduation between the three mode groups to see whether older alumni were less likely to complete a survey on the Web or through IVR. Neither version of the age variable showed a significant difference by mode.

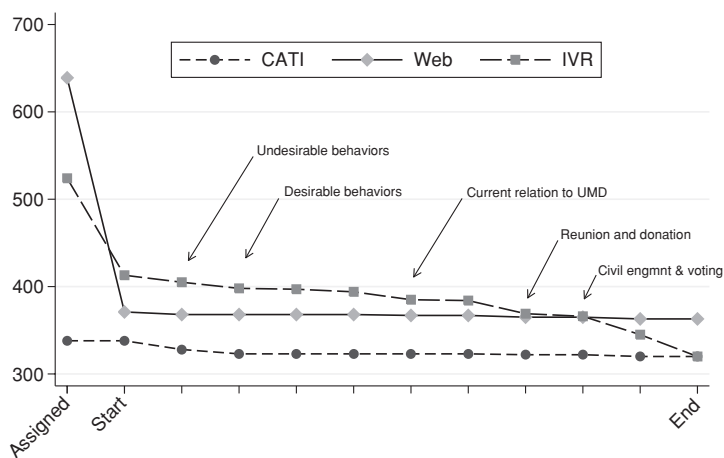


Figure 1. Distribution of Dropouts by Mode: Number of Respondents Responding at 10 Time Points between Random Assignment, Start of the Questionnaire, and the End of the Survey.

NOTE.—Question sections are marked inside the graph.

poor relationships with the University were somewhat less likely to complete the main questionnaire but the overall biases were small and did not differ by mode. It is conceivable that the reports of the nonrespondents would have been affected by mode differently from those of respondents; unfortunately, this conjecture cannot be tested with the data at hand.

Analysis

We present the analysis in three parts, each related to our initial research questions. First, we compare the survey reports by mode without making use of the data from official University records. Next, we examine accuracy of the survey reports by mode, comparing the survey reports to the University records. Both of these analyses shed light on the relative effectiveness of the two self-administered modes (IVR and Web) for eliciting sensitive information compared to an interviewer-administered telephone interview. Our final set of analyses explore whether the mode of data collection affects perceived question sensitivity.

DIFFERENCES IN REPORTING AND ITEM NONRESPONSE BY MODE

Table 6 shows the proportion of respondents reporting each of the undesirable and desirable characteristics by mode of data collection. In addition, it shows

Table 6. Proportions Reporting Desirable and Undesirable Characteristics, by Item and Mode of Data Collection

	CATI	IVR	Web	ANOVA/ χ^2 (df; <i>p</i> -value)	<i>t</i> -statistic/ χ^2 (df; <i>p</i> -value), CATI versus Web
Percentage “yes” summing across the four undesirable characteristics	25.8	27.7	30.5	$F(2, 996) = 2.72, p = .07$	$t = -2.30 (680; .01)$
GPA < 2.5	1.8	3.7	6.2	7.74 (2; .021)	7.30 (1; .01)
At least one D or F	42.2	44.3	50.7	5.44 (2; .066)	4.90 (1; .03)
Dropped a class	46.7	45.6	50.6	1.84 (2; .399)	0.99 (1; .32)
Warning/probation	10.2	13.4	13.8	2.34 (2; .311)	2.07 (1; .15)
Percentage “yes” summing across the five desirable characteristics	27.0	25.5	26.0	$F(2, 997) = 0.27, p = .76$	$t = 0.554 (688; .71)$
GPA > 3.5	23.8	20.4	24.2	1.43 (2; .489)	0.01 (1; .92)
Honors	16.3	19.9	15.5	2.58 (2; .275)	0.08 (1; .77)
Ever donate	42.1	40.5	41.3	0.15 (2; .927)	0.04 (1; .84)
Donated in last year	44.2	41.9	40.5	0.36 (2; .835)	0.36 (1; .55)
Member of Alumni Association	24.8	21.5	23.6	0.97 (2; .615)	0.12 (1; .73)

Table 7. Item Nonresponse Rates, by Mode of Data Collection

	Item missing (in percent)		
	CATI	IVR	Web
Undesirable			
At least one D or F	2.19	1.88	0.00
Dropped a class	6.25	3.44	0.28
Warning/probation	1.56	1.88	0.00
Desirable			
Honors	4.06	2.81	0.28
Ever donated	3.44	4.38	0.55
Member of Alumni Association	2.81	2.50	0.83
GPA ^a	12.80	15.60	1.93

NOTE.—^aThe missing item rates for GPA are reported overall, not grouped by low and high GPA.

two summary measures, one for the four undesirable items and one for the five desirable times.

We examined the proportion of the four undesirable items that respondents answered affirmatively. On the average, this was lowest for the interviewer-administered telephone interview (CATI showed an average of 25.8 percent) and highest for the Web (30.5 percent). The corresponding figure for IVR was 27.7 percent. A one-way ANOVA among the three modes shows that these differences are marginally significant ($p < .07$), with the difference between CATI and Web reaching significance ($p < .02$). This overall pattern generally holds at the item level. The rate of positive reporting for all four of the undesirable items is highest via the Web but the difference by mode is significant only for low GPA ($p < .05$); it is marginally significant for grades of D or F ($p < .07$).

In contrast, for the socially desirable items, there are no significant differences across modes either for the average proportion of “yes” answers to the five desirable items or for any of the individual items.

Another way by which respondents can avoid making embarrassing admissions about themselves is to skip the question. Table 7 shows the item nonresponse rates by mode on our key questions. (We leave out donations in the prior year, since that item was skipped for the respondents who said that they had never made a donation to the University). The CATI and IVR respondents were more likely not to answer these questions than the Web respondents.⁹ Most of the CATI item nonresponse comes from respondents saying they didn’t know instead of refusing to answer. The distinction between don’t know and refusal is not available for the IVR and Web mode.

9. With the exception of Alumni Association membership, all of these differences show significant chi-square values ($p < .05$).

Table 8. Percent Reporting Having a D or F for a Class, by Recorded Status

		Survey Report of D or F		
		Yes	No	Total
Status according to records	Yes	73.4	26.6	60.7
	No	3.6	96.4	39.3
Total		46.0	54.0	<i>n</i> = 990

ACCURACY OF REPORTING BY MODE

Because we have external record data, we can compare the modes not only on levels of reporting (and levels of item nonresponse) but on levels of accuracy as well. By comparing the survey reports to the record data, we can measure two types of errors: (1) behavior that is reported in the survey but not in the records represents a false positive or overreport, while (2) behavior that is not reported in the survey but is indicated in the records represents a false negative or underreport.

As an example, table 8 shows the false positive and false negative rates for grades of D or F. Of the 990 respondents who answered that question, 60.7 percent had received such a grade according to the records. Of those, roughly 27 percent did not report an unsatisfactory grade (false negatives). The false positive rate (respondents claiming a grade of “D or F” when records show they did not receive one) is only about 4 percent. We exclude respondents who didn’t answer the question from this analysis and the analysis of other individual items.

Table 9 shows false negative and false positive rates for each of the items by mode. The errors in the expected direction are boldfaced. For the three socially undesirable items, the false negative rates are much higher than the false positive rates and this is true for all three modes of data collection. For all three socially undesirable items, the false negative rates are lowest for the Web respondents. For the socially desirable items, the difference between the false negative and false positive rates across all survey modes is less pronounced and the differences across the modes are somewhat smaller.

Some of the false negative and false positive rates are based on small numbers of cases. For example, only about 2 percent of the respondents actually received a warning or were placed on academic probation; only 12 percent received honors. For four of the items, more than half of the respondents in each mode group were in the socially undesirable category (that is, they had in fact received at least one D or F, dropped a course, earned a GPA less than 3.5, or never donated money to the University). We carried out tests examining mode differences for these four individual items and found that the mode differences were significant for two of them (the item asking about getting a D or an F and the item on the respondent’s GPA).

Table 9. False Negative and False Positive Rates, by Item and Mode

True status	CATI		IVR		Web		All respondents
	False negative (%)	False positive (%)	False negative (%)	False positive (%)	False negative (%)	False positive (%)	Yes (record) (%)
GPA < 2.5	83.3	0.0	69.2	0.8	61.5	0.7	12.1
At least one D or F	33.0	3.3	28.3	5.4	19.9	2.2	61.1
Dropped a class	34.3	6.3	34.2	9.1	31.6	7.6	68.4
Warning or probation	33.3	9.1	33.3	11.8	25.0	12.4	2.4
GPA > 3.5	16.7	7.4	19.1	1.9	6.8	6.0	21.4
Honors	5.3	5.2	0.0	5.7	2.8	6.4	12.1
Donations ever	31.5	24.3	25.4	19.2	30.5	20.3	40.7
Donations in last year	8.8	25.6	22.2	25.9	20.0	23.3	15.7
Member of Alumni Association	2.0	10.7	11.1	10.1	3.2	8.1	16.2

To boost power and examine the pattern of results across the entire set of items, we pooled across all seven items. The pooled analysis examined the false negative rates for the items involving socially undesirable behaviors and the false positive rates for the items involving socially desirable behaviors. We fitted a random-effects logit model that treated the data as clustered by respondent; it estimated the mode differences taking into account all of the items for which record data were available. Compared to CATI respondents, Web respondents were significantly less likely to misreport in a socially desirable direction (odds ratio of 0.8). The difference between IVR and CATI is somewhat smaller and marginally significant (odds ratio of 0.86; $p = .13$). The misreporting rates for the Web and IVR do not differ significantly ($p > .50$). Testing for the overall difference between IVR and Web versus CATI gives an odds ratio of 1.21, reflecting CATI's significantly higher rate of misreporting ($p < .05$).¹⁰

RATINGS OF QUESTION SENSITIVITY

One possible explanation for the mode differences in social desirability bias in tables 6 and 9 is the impact of mode of data collection on the sensitivity of the question. Question sensitivity is likely to reflect the true status of the respondents but may also vary by mode. That is, the questions probably seem less sensitive to those in the desirable category than to those in the undesirable category and they may also seem less sensitive when a computer administers the questions than when an interviewer does. We examine these possibilities with the responses to the items at the end of the questionnaire that asked respondents whether they thought the first four questions listed in table 1 might make people they know report falsely or exaggerate their answers.

Figure 2 shows the average perceived sensitivity of each item by the true status of the respondent and the mode of data collection. For example, 41 percent of the CATI respondents who had in fact dropped a class considered the question about dropping a class as one for which people they knew were likely to misreport. For all four items (and regardless of whether the respondents were in the socially desirable or the socially undesirable category), the CATI respondents were most likely to say that people might not answer the questions truthfully. With the exception of "Dropping a class," respondents in the socially undesirable category were more likely to perceive an item as sensitive.

The results from a logistic regression with perceived sensitivity as the dependent variable are given in table 10. They confirm the patterns apparent in figure 2. For all four items, mode significantly affected the perceived sensitivity of the items; perceived sensitivity was lower for both self-administered modes than for CATI (demonstrated by the odds ratios less than 1). In addition, for three of the four items, there was a significant effect for the respondent's

10. Adding an indicator variable for recent alumni did not change the odds ratio of IVR and Web.

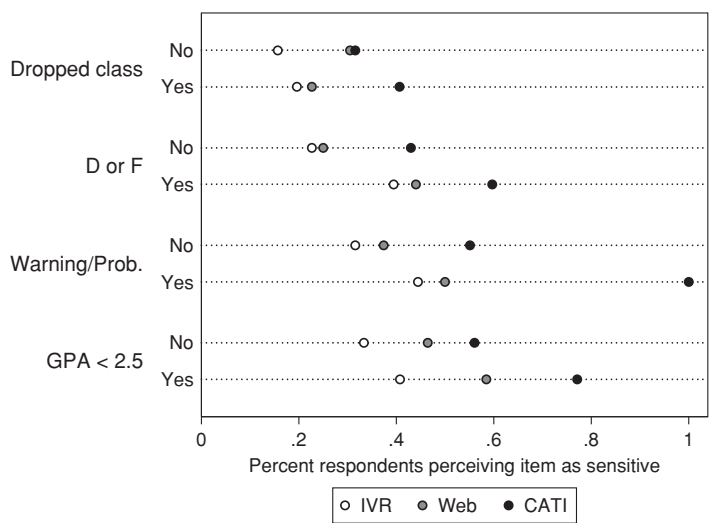


Figure 2. Mean Sensitivity Ratings, by Item, Mode, and True Status.
NOTE.—For each item, the top row represents the desirable state and the bottom row the undesirable state.

Table 10. Odds Ratios (and Standard Errors) From Logit Models for Perceived Sensitivity by Mode and Status

	Dropped class	D or F	Warning/probation	Low GPA
Web	0.546 (0.093)	0.498 (0.081)	0.471 (0.075)	0.660 (0.104)
IVR	0.367 (0.071)	0.420 (0.073)	0.363 (0.062)	0.374 (0.064)
Undesirable category	1.071 (0.172)	2.172 (0.311)	2.560 (1.114)	1.775 (0.366)
<i>n</i>	951	955	956	954
Likelihood ratio χ^2	29.83	60.20	44.43	43.27
Prob. > χ^2	.000	.000	.000	.000

actual status—those in the undesirable category found the items significantly more sensitive (demonstrated by odds ratios greater than 1) than those in the desirable category. Even with the one item for which true status didn’t have a significant effect (the item on having dropped a course), the trend was in the right direction.

Discussion

We report three main findings, corresponding to the research questions we posed initially. First, Web administration increased the reporting of sensitive

information relative to conventional CATI, with IVR intermediate between the other two modes (table 6). These differences by mode were larger for the socially undesirable items than for the socially desirable ones. Researchers have speculated that Web surveys might produce increases in reporting of sensitive information similar to those of other methods of self-administration but ours is the first study we are aware of that demonstrates such gains (Knapp and Kirk 2003 found no differences between modes in a comparison of the Web, IVR, and paper self-administration¹¹).

Second, the increased levels of reporting in the Web represented increased accuracy. Not only were Web respondents more likely than CATI respondents to report more socially undesirable things about themselves, they were less likely to falsely deny them (lower false negative rates for the undesirable items); IVR was generally in between the other two modes (table 9). Although IVR has figured in several mode studies (mainly with CATI as the comparison group), ours is the first that included validation data. As with the differences in reporting, the improvements in accuracy produced by the Web (and to a lesser extent by IVR) were much more apparent for the items concerning undesirable characteristics than for those about desirable characteristics. The preponderance of the errors was in the socially desirable direction for all but one of the seven items for which we had validation data. The one exception (which we cannot explain) was the question on ever having donated to the Alumni Association. This behavior tended to be reported poorly overall, with underreporting more likely than overreporting, regardless of the respondent's true status.

Finally, both mode of data collection and the actual status of the respondent influenced whether respondents found an item sensitive as measured by our question about whether other people might misreport or exaggerate in responding to the item. For all four items, the IVR respondents were least likely and CATI respondents most likely to say that people would misreport. In addition, for all of the items and all three modes of data collection, the respondents in the socially undesirable category were more likely than those in the socially desirable category to say that people would misreport. As might be expected, there was a substantial variation in sensitivity across the four items, with the question on GPA deemed the most sensitive of the four items. Somewhat surprisingly, items were seen as more sensitive by the Web respondents than by the IVR respondents, even though the Web respondents answered them more accurately and were less likely to skip them than the IVR respondents. This may reflect lingering concerns among some of the respondents about whether data that are transmitted over the Web are really secure.

Although we found that differences by mode in reporting were much clearer for the questions about undesirable characteristics than for those about desirable

11. One should note that the experimental groups in Knapp and Kirk (2003) are very small and many of their tests have little power to detect significant results. The study also did not include any interviewer-administered condition.

ones (see tables 6 and 9), this might be a function of the specific items we used. A question about getting a grade of a D or an F, for example, could be more sensitive than one about graduating with honors and therefore more prone to misreporting. With other items, the pattern might differ.

Our study provides a good illustration of both the value and limitations of using validation data to study reporting error for questions that raise social desirability issues. Records data allowed us to conduct more sensitive tests of mode differences in reporting, to determine whether the differences in reporting that we observed represented differences in accuracy, and finally to examine whether nonresponse affected comparisons across the different mode groups. Still, our ability to access these data involved a cost. Our study involves a specialized population—recent alumni of a single university—and focuses on questions about a specialized set of topics relevant to that population.

We suspect that our findings are likely to generalize to other questions and to other populations. Even with our not-very-sensitive questions, we find mode differences in both levels of reporting and reporting accuracy that are consistent with the past literature on social desirability biases and mode effects (Tourangeau and Yan 2007). We note, however, that our sample was relatively young and well educated and therefore might have been more receptive to computer administration of the questions than members of the general population. Social desirability bias is, of course, not the only source of reporting error to these questions. Time since graduation, via recall error, contributed as well. But although older alumni (those who graduated before 1995) misreported more than younger ones, the effects of mode were the same for the two groups.

A possible weakness of our study is that identifying the survey sponsor as the University of Maryland might have led some respondents to suppose that the researchers would have access to the validation variables. Although this might have reduced the overall level of misreporting, we have no reason to believe that it would have operated differently across mode and thus do not believe that it affected our mode comparisons.

Considering all our findings, no mode of data collection dominated the other two. Each of our three main outcome variables—unit nonresponse, item nonresponse, and reporting accuracy—yielded a different ranking of the modes. CATI had the best response rate and the Web, the worst. CATI had the highest rate of item missing data and the Web the lowest. The Web had the highest levels of reporting accuracy and CATI had the worst. Thus the choice of mode could depend on which source of error is most important for a survey.

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